In [5]:
```
import pandas as pd
import numpy as np
```

In [6]:
```
df = pd.read_csv("train (3).csv")
df.head()
```

Out[6]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500

In [7]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   PassengerId  891 non-null    int64  
 1   Survived     891 non-null    int64  
 2   Pclass       891 non-null    int64  
 3   Name         891 non-null    object 
 4   Sex          891 non-null    object 
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64  
 7   Parch        891 non-null    int64  
 8   Ticket       891 non-null    object 
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object 
 11  Embarked     889 non-null    object 
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

In [10]:
```
df.shape
```

Out[10]:
```
(891, 12)
```

In [11]:
```
df.describe
```

Out[11]:
```
<bound method NDFrame.describe of      PassengerId  Survived  Pclass  \
0              1         0       3
1              2         1       1
2              3         1       3
3              4         1       1
4              5         0       3
..           ...       ...     ...
886          887         0       2
887          888         1       1
888          889         0       3
889          890         1       1
890          891         0       3

                                                Name     Sex   Age  SibSp  \
0                             Braund, Mr. Owen Harris    male  22.0      1
1    Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1
2                              Heikkinen, Miss. Laina  female  26.0      0
3         Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1
4                           Allen, Mr. William Henry    male  35.0      0
..                                                ...     ...   ...    ...
886                            Montvila, Rev. Juozas    male  27.0      0
887                     Graham, Miss. Margaret Edith  female  19.0      0
888          Johnston, Miss. Catherine Helen "Carrie"  female   NaN      1
889                            Behr, Mr. Karl Howell    male  26.0      0
890                              Dooley, Mr. Patrick    male  32.0      0

     Parch            Ticket     Fare Cabin Embarked
0        0         A/5 21171   7.2500   NaN        S
1        0          PC 17599  71.2833   C85        C
2        0  STON/O2. 3101282   7.9250   NaN        S
3        0            113803  53.1000  C123        S
4        0            373450   8.0500   NaN        S
..     ...               ...      ...   ...      ...
886      0            211536  13.0000   NaN        S
887      0            112053  30.0000   B42        S
888      2        W./C. 6607  23.4500   NaN        S
889      0            111369  30.0000  C148        C
890      0            370376   7.7500   NaN        Q

[891 rows x 12 columns]>
```

In [14]:
```
#mean
df.mean(numeric_only=True)
```

Out[14]:
```
PassengerId    446.000000
Survived         0.383838
Pclass           2.308642
Age             29.699118
SibSp            0.523008
Parch            0.381594
Fare            32.204208
dtype: float64
```

In [15]:
```
#Median
df.median(numeric_only=True)
```

Out[15]:
```
PassengerId    446.0000
Survived         0.0000
Pclass           3.0000
Age             28.0000
SibSp            0.0000
Parch            0.0000
Fare            14.4542
dtype: float64
```

In [17]:
```
#Mode
df.mode()
```

Out[17]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	C
0	1	0.0	3.0	Abbing, Mr. Anthony	male	24.0	0.0	0.0	1601	8.05	
1	2	NaN	NaN	Abbott, Mr. Rossmore Edward	NaN	NaN	NaN	NaN	347082	NaN	
2	3	NaN	NaN	Abbott, Mrs. Stanton (Rosa Hunt)	NaN	NaN	NaN	NaN	CA. 2343	NaN	
3	4	NaN	NaN	Abelson, Mr. Samuel	NaN	NaN	NaN	NaN	NaN	NaN	
4	5	NaN	NaN	Abelson, Mrs. Samuel (Hannah Wizosky)	NaN	NaN	NaN	NaN	NaN	NaN	
...	...	...	...	...	...	...	...	...	...	...	
886	887	NaN	NaN	de Mulder, Mr. Theodore	NaN	NaN	NaN	NaN	NaN	NaN	
887	888	NaN	NaN	de Pelsmaeker, Mr. Alfons	NaN	NaN	NaN	NaN	NaN	NaN	
888	889	NaN	NaN	del Carlo, Mr. Sebastiano	NaN	NaN	NaN	NaN	NaN	NaN	
889	890	NaN	NaN	van Billiard, Mr. Austin Blyler	NaN	NaN	NaN	NaN	NaN	NaN	
890	891	NaN	NaN	van Melkebeke, Mr. Philemon	NaN	NaN	NaN	NaN	NaN	NaN	

891 rows × 12 columns

In [18]:
```
df.dtypes
```

Out[18]:
```
PassengerId      int64
Survived         int64
Pclass           int64
Name            object
Sex             object
Age            float64
SibSp            int64
Parch            int64
Ticket          object
Fare           float64
Cabin           object
Embarked        object
dtype: object
```

In [19]:
```
df.groupby('Sex')['Age'].mean()
```

Out[19]:
```
Sex
female    27.915709
male      30.726645
Name: Age, dtype: float64
```

In [20]:
```
df.groupby('Pclass')['Fare'].mean()
```

Out[20]:
```
Pclass
1    84.154687
2    20.662183
3    13.675550
Name: Fare, dtype: float64
```

In [21]:
```
df.isnull().sum()
```

Out[21]:
```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

In [24]:
```
df['Age'] = df['Age'].fillna(df['Age'].mean())
```

In [25]:
```
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])
```

In [26]:
```
df.isnull().sum()
```

Out[26]:
```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age              0
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         0
dtype: int64
```

In [29]:
```
#Convert Categorical Data to Numeric
from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()
df['Sex'] = le.fit_transform(df['Sex'])
```

In [31]:
```
df.columns
```

Out[31]:
```
Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp',
       'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked_Q', 'Embarked_S'],
      dtype='object')
```

In [35]:
```
if 'Embarked' in df.columns:
    df = pd.get_dummies(df, columns=['Embarked'], drop_first=True)
```

In [36]:
```
from sklearn.preprocessing import OneHotEncoder

enc = OneHotEncoder()
enc_df = pd.DataFrame(enc.fit_transform(df[['Pclass']]).toarray())
enc_df.head()
```

Out[36]:

	0	1	2
0	0.0	0.0	1.0
1	1.0	0.0	0.0
2	0.0	0.0	1.0
3	1.0	0.0	0.0
4	0.0	0.0	1.0

In [37]:
```
df_encoded = df.join(enc_df)
df_encoded.head()
```

Out[37]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare
0	1	0	3	Braund, Mr. Owen Harris	1	22.0	1	0	A/5 21171	7.2500
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	0	38.0	1	0	PC 17599	71.2833
2	3	1	3	Heikkinen, Miss. Laina	0	26.0	0	0	STON/O2. 3101282	7.9250
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	0	35.0	1	0	113803	53.1000
4	5	0	3	Allen, Mr. William Henry	1	35.0	0	0	373450	8.0500

In [38]:

```
df.min(numeric_only=True)
df.max(numeric_only=True)
```

Out[38]:

```
PassengerId         891
Survived              1
Pclass                3
Sex                   1
Age                80.0
SibSp                 8
Parch                 6
Fare           512.3292
Embarked_Q         True
Embarked_S         True
dtype: object
```

In [ ]: